Words, Color, and Events

console.log() — a window only you can see

```
console.log(landerY);
console.log("got here");
```

It prints underneath the editor, in the **console** — not on the canvas. Nobody using your program ever sees it.

> **The console is for you. The canvas is for the user.** Both put a value where it can be read; only one is meant to be seen by anybody else.

Reach for it when

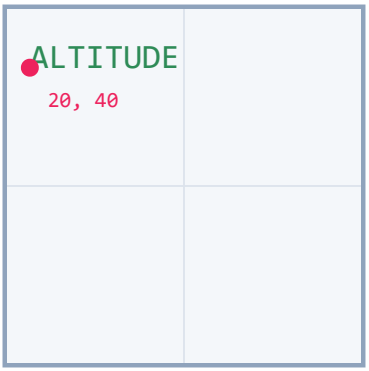
SYMPTOM	QUESTION IT ANSWERS
nothing moves	is the variable actually changing?
nothing happens	is this line running at all?
it looks wrong	what is in there, right now?

Inside `draw()` it prints about sixty times a second, so the console fills fast. That is normal — clearing it does not stop it.

text()

```
fill(120, 220, 140);
textSize(24);
text("ALTITUDE", 20, 40);
```

`text(what, x, y)` — what to show, then where. `fill()` sets its color, same as any shape.



> **The y is the bottom of the letters**, not the top. At `y = 0` the words are drawn above the canvas and you see nothing.

Strings — what quotes do

```
text(landerY, 20, 40);   // 210
text("landerY", 20, 80); // landerY
```

Without quotes it is a **variable** and you get what is inside it. With quotes it is a **string** — just those characters.

Joining them

```
text("ALT " + landerY, 20, 40);
```

The `+` glues a string to a number. That is called **concatenation**.

> `"200"` and `200` are different things. One is characters, one is a number. `"200" + 1` gives `"2001"`, because for strings `+` means *join*, not *add*.

Colors in variables

```
let hullColor = color(214, 222, 235);
fill(hullColor);
```

`color()` takes the same numbers as `fill()` — three for RGB, one for gray, four with alpha. Storing it means you change the color in one place instead of six.

Code that runs when something happens

```
function keyPressed() {
  landerY = 40;
}
```

runs **once**, when a key goes down

```
function mousePressed() {
  speed = 0;
}
```

runs **once**, when the mouse is clicked

> **Three ways these silently do nothing.** No error message for any of them.
> - Written *inside* `draw()` instead of beside it
> - Spelled `keypressed` — the capital P matters
> - Misspelled any other way — p5 looks for that exact name

> **Why one line can reset a whole drawing.** Every part of the rocket is written as an offset from `landerY`, so putting that variable back puts everything back. A drawing built from fixed numbers could not be reset at all.

On the AP exam

What you write

```
text("ALT " + landerY, 20, 40);
```

Exam Reference Sheet

```
DISPLAY(landerY)
```

> The exam has no canvas, so `DISPLAY` takes one thing and shows it — no x, no y. Exam questions often end with a `DISPLAY` and ask what appears.

Where this fits in AP CSP

CODE	LEARNING OBJECTIVE
`3.1`	Variables and Assignments
`3.4`	Strings
`AAP-1.A`	Represent a value with a variable
`AAP-2.D`	Evaluate expressions that manipulate strings
`4.B`	*Skill:* determine the result of code segments

The types so far. The exam uses numbers, Booleans, lists and strings. You now have **numbers** and **strings**. Booleans arrive next lesson. `color` is p5's own type, not one of the exam's.
Two ways to see a value. `console.log()` for yourself while you are working; `text()` for whoever uses the program.